

PENNDOT TOOLBOX for DEVELOPMENT of a WAYFINDING SIGNING REGION

I. Purpose of this Toolbox

The information included in the toolbox has been provided to explain the steps necessary for a geographic region to become a Signing Region. Tourism signing will likely be more effective if the signs are installed and maintained at a local regional level, rather than at a statewide level, as it is today.

Using this toolbox a Signing Region can establish a systematic network of traffic directional signs (Wayfinding Signs) in order to provide a sequence of “intermediate” geographical steps to guide motorists from major highway gateways to most commercial, civic and cultural/visitor destinations in the region, and then back out again. The goal of such a program is to provide comprehensive guidance assistance with a minimum of signs by first using Wayfinding Signs to “make sense” of how a region is structured. Directional signs for individual destinations are still permitted under a Regional program, but only as the last directional “step”. Thematic corridor signing is also permitted, but only as long as these specially marked corridors coincide with a natural travel corridor and work as one of the Region’s “geographical steps”.

II. Background

For information purposes, a brief description of the Tourist Oriented Destination Signing Policy (TODS), General Motorist Services (GMS) signs, the PA Logo Signing Program and Wayfinding Signs are described below:

A. TODS

The purpose of the TODS policy is to establish guidelines for the installation of signs within State highway right-of-way to guide travelers to businesses, services, and attractions in which the traveling public would have reasonable interest. These guidelines include the eligibility, location, design, installation, cost, and maintenance of these signs, which hereinafter will be referred to as "TODS." These signs are for individual attractions, and are approved through PennDOT, or the administrator of a signing district. TODS are not authorized for use on a freeway.

B. GMS

PennDOT provides General Motorist Service (GMS) signs to indicate the availability of gas, food, lodging, camping, visitor information, hospitals, and State Police, and information at freeway interchanges. By definition, gas, food, lodging, and camping facilities must be within one, two, three, and five miles,

respectively, from the end of the exit ramp. GMS signs provide generic information for motorists but no information for tourists desiring to visit attractions in the area. GMS signs are provided at no cost to the businesses.

C. Logos

The PA Logo Signing Program was developed by PennDOT and is administered by the Pennsylvania Travel Council. This program uses special motorist service signs with the logos of businesses to highlight the specific businesses which offer gas, food, lodging, camping, and attractions.

D. Wayfinding

Wayfinding Signing is a systematic network of directional signs to guide the traveling public from major highway gateways to key civic, cultural, visitor, recreational and commercial destinations within a specified Signing Region with the minimum number of signs.

III. Wayfinding Signs

A. Signing Districts and Regions

Signing Districts are generally confined to one municipality and administered by a single governmental sponsor. A Signing Region is a geographical area with defined boundaries including multiple municipalities which has substantial, significant and varied tourism facilities and opportunities that collectively provide a tourism experience reasonably unique to the region. At a minimum, a region should include an entire county or the majority portion of at least two counties and include a major highway corridor facility identified in the Pennsylvania Statewide Long Range Transportation Plan (2000 – 2025). The advantage of a Signing Region is that it will allow signs for an attraction being posted outside of that attraction's municipality. Generally, no new Signing Districts should be established within an approved Signing Region to eliminate potential confusion to motorists.

B. Sign Types

The goal of a Wayfinding Sign system is to take advantage of known geographical sequences to replace turn by turn, street by street driving directions with short, easy to remember geographical narratives that gets visitors to within a few blocks of any destination. Wayfinding Signs should be used as tools to create sequences of predictive geographical steps that guide visitors from every highway gateway into any participating destination and then back out again. In the process, a Region's most prominent features and assets are called out and promoted, but only as required to serve visitor guidance. In general, individual attractions receive far fewer signs in this kind of navigational sign system than

they would have received under the TODS program of individual first-come, first-served signing. But the community and the attractions gain a comprehensive and consistent system of signing that makes the whole region much more accessible to the visitor while keeping sign load and the potential for confusion to a minimum.

Signing regions should include a Region Logo, Welcome Signs, Guideway Markers, Region Trailblazer Signs, and Resource Directional Signs, described as follows:

Region Logo – A logo unique to the region which is incorporated into various signs in the Wayfinding System. (See Figure 1.15.1)

Welcome Sign – A sign with the name of the region placed on arterials at the boundary of the region. (See Figure 1.15.2 – 4)

Region Trailblazer Sign – A Sign placed outside of the region directing motorists towards a region. (See Figure 1.15.5)

Guideway Marker – A sign used to designate a travel corridor spanning as a guideway. (See Figure 1.15.6 - 7)

Resource Directional Sign - A sign with the named or grouped attraction indicating the direction (and sometimes the distance) to that attraction or group of attractions. Up to three attractions may be posted on each structure, with the attractions being separate modular signs. The sign structure should remain in place if individual attractions change. (See Figure 1.15.8 - 10)

Signing Regions may, where necessary, include sub-regions in Urban Areas. Sub-regional signs are summarized as follows:

Sub-region Resource Directional Sign – A sign with named or grouped attractions indicating the direction (and sometimes distance) to the attraction or group of attractions within the sub-region. (See Figure 1.15.11)

Sub-region Direction Sign – A sign used to provide directions from one sub-region to another sub-region. (See Figure 1.15.12)

Region Direction Sign – A sign used to provide directions from a sub-region back into the greater region. (See Figure 1.15.13)

C. Treatment of Existing TODS Outside of a Signing Region

The TODS program should continue under the current Department policy with the following modification. All new agreements for TODS should include a provision that the TODS may be removed when a Signing Region is established,

that includes the location of the TODS installation regardless of whether the attraction participates in the Regional Signing program.

D. Treatment of Existing TODS Inside of a Signing Region

1. No new TODS installations should be approved upon the establishment of a Signing Region.
2. Existing TODS of attractions that participate in the Regional Signing program shall be removed.
3. TODS participants that elect not to participate in the Signing Region will only be authorized to keep their TODS for 5 years from the date of the establishment of the Signing Region. After that time their TODS will be removed. If in the interim their TODS is damaged, they may elect to replace it at the price in effect at the time of replacement or join the Signing Region if they qualify and space is available. If replaced, it will still remain subject to removal after the 5 year period.
4. Existing TODS that conflict with the installation of a Regional sign installation shall be relocated to an alternate location. In the case that an alternate location cannot be identified, the Regional Trust should work with the business to find a mutually acceptable resolution
4. Existing TODS that serve as trailblazer signs in the Pa Logo Signing Program should be replaced with signs that conform to the Regional Signing Program design. The TODS replacements shall be coordinated with the Administrator of the Pa Logo Signing Program.

IV. Developing a Signing Region

A. Stakeholder Development

The group interested in creating a Signing Region should initially form a group of stakeholders, consisting of a representation of diversified interests that could potentially be involved in the signing region. Stakeholders should include as appropriate (but not be limited to) representatives of the following groups:

- Potential participants (attractions desiring signing)
- Municipalities and Counties
- County Planning Commissions
- Vacation, Tourism and/or Visitor Bureaus

- PennDOT Engineering District and Central Offices
- Pennsylvania Department of Community and Economic Development (PDCED)
- Pennsylvania Historical & Museum Commission (PHMC)
- Pennsylvania Travel Council
- Pennsylvania State Parks – Department of Conservation and Natural Resources
- Federal Attractions/Parks (National Park Service; U.S. Forest Service; U.S. Army Corps of Engineers; etc.)

It is recommended that the stakeholders retain a qualified technical consultant with experience in Wayfinding Sign Systems to assist in the formation of the Signing Region.

B. Preliminary Plan Process

After the group has been formed, the stakeholders should define the size and boundaries of the signing region with logical borders considering its unifying characteristics and generally utilizing county or municipal lines. The stakeholders begin this planning process by identifying on a map of the region the grid of arterials [the designated arterials] and Guideways (to be specially marked with Guideway route markers) which will serve as the starting point for signing for eligible destinations. Since most destinations in rural areas will be eligible to begin signing only from the designated arterial or Guideway nearest their locations, this grid should encompass only travel routes that are likely to have visitor traffic.

If there is an adjacent signing region established, there must be cooperation and interaction between the established and proposed regions, with appropriate placement of Region Trailblazer signs. Similar signing from one region to another will enable tourists travelling between regions to readily recognize the attraction signs.

The stakeholders' Preliminary Plan submission to PennDOT should include:

1. Indicate what existing attractions in the region might desire signing (the potential participants), and

Utilizing existing data sources, including directories and listings from TPAs, DCED, Chambers of Commerce, state parks, economic development agencies, etc., provide a listing of potential participants including contact information.

Correlate this listing to a key map showing approximate locations of attractions.

2. Identify Attractions of Regional Significance.

The stakeholders should designate up to approximately twelve (12) “attractions of regional significance” [RS-Attractions]. The number of attractions of regional significance should not be more than 2.5% of the number of potential participants. RS-Attractions would include the largest commercial traffic generators (among eligible categories of destinations) and the most significant and region-defining cultural, historic, and civic assets. These attractions provide definition to the region due to their noteworthiness and name recognition outside of the region. The stakeholders can then use signs pointing to the RS-Attractions as regional “lighthouses” as it were, using them to guide visitors to the “lesser” destinations in the vicinity of each of the “lighthouses”. At the junction of arterials and/or Guideways, signs pointing to RS-Attractions can also be used to provide a regional sense of orientation as to which way is which. Thus, the stakeholders should select RS-Attractions both for their prominence and for how their geographic distribution provides for regional guidance.

3. Identify major travel routes leading to the region and existing traffic patterns for motorists traveling to and within the region and potential attractions.

The major travel routes leading to the region should be identified by reviewing PennDOT Functional Classification Maps, with Interstates, Freeways and Arterials being highlighted on said map. Existing traffic patterns for motorists traveling to and within the region and potential attractions should be identified based on input from PennDOT District Traffic Unit staff, DCED, TPA and stakeholders. These travel patterns should be highlighted on a map.

4. Identify the existing wayfinding or attraction signs in the proposed region.

Utilizing available information from PennDOT records, adjacent signing regions, if applicable, and field views of the numbered travel routes in the region, prepare an exhibit showing the type and location of existing attraction, TODS, and Wayfinding signs.

5. Identify Potential Guideways.

The stakeholders can create a region-sized travel route called a “Guideway”. Guideways are designated travel corridors spanning a distinct geographic area. Guideway routes are identified by route marker signs displaying a unique name predictive of the geographical area that it spans or of the entities located along it. These route markers would not replace existing street names or state

route designations. They can, in fact, complement them, as in the case of the “National Road” for US 40 and the “Lincoln Highway” for US 30. Note in the conditions of use below that Guideways must be constructed carefully so that they actually serve to improve guidance.

There are two reasons for creating marked Guideways:

- a. A natural travel corridor of regional significance is obscured by topography or by overlapping or changing state route numbers and/or street names; or
- b. A natural travel corridor spans an area that is dominated with destinations of a certain type or visitor theme (a heritage corridor, for example, or a “780 Research Corridor”), and adding Guideway route markings to the corridor will aid visitors in finding this class of destinations and reduce the need for individual Resource Directional Signs.

Note that Guideways are not “paths” or “loops” or “trails” laid out simply to “connect the dots” from one destination to the next. They are used as a tool for articulating the structure of a region, one of the geographical steps that can lead to the signed trail for an individual destination.

6. Identify urban areas.

Urban areas shall be identified based on PennDOT County Functional Classification Maps.

7. Identify potential Sub-regions.

In officially designated urban areas, the stakeholders can create up to five sub-regions in urban areas as the intermediate geographical step to locating destinations. Each sub-region should constitute a neighborhood or district with a shared “sense of place”.

Sub-regional signing would be accomplished in the manner described below. Potential sub-regions should be identified on a map of the region. The stakeholders would identify attractions of “sub-regional significance” [sRS-Attraction] that would be the only entities eligible for signing along designated arterials and Guideways, and then only within their respective sub-region. A lesser category of sRS-Attraction can also be identified by the stakeholders which would be restricted to signing as per the following.

Guidance among sub-regions is organized into three stages:

- a. At the edges of sub-regions on designated arteries and Guideways, signs point the way to the neighboring sub-regions as well as to the major

highways for exiting to neighboring regions [Sub-region and Region Signs];

- b. When the neighboring sub-region is reached, the Sub-region Signs lead onto a designated artery or Guideway off of which every local sRS-Attraction is then signed with Resource Directionals;
- c. Attractions of lesser importance may then be signed from one of the sRS-Attractions offshoot routes, also on Resource Directionals.

In this way, an unbroken chain of guide signs can be created to every participating attraction based on no more than two intermediating geographical steps (with visitors needing to know the sub-region and then the sRS-Attraction with which a lesser attraction is associated). And with the use of supplemental signing which advises motorists as to the type of destinations to be found in a sub-region or by the careful selection of sub-region names that distinguish the kinds of destinations to be found within them, visitors may get all they need to know from the signs themselves.

The stakeholders should submit this Preliminary Plan to PennDOT for review and approval. The stakeholders may appoint an individual (which may be a stakeholder) or hire a consultant to act as the contact person with PennDOT on behalf of the stakeholders. The contact person should be familiar with PennDOT and MUTCD sign standards and engineering practices. Stakeholders must reach agreement (consensus) on the above items as evidenced by Letters of Support prior to submitting the Preliminary Plan to PennDOT.

C. Final Plan Process

After PennDOT approves the Preliminary Plan for a Signing Region, the stakeholders of the Signing Region will develop a Final Plan with PennDOT acting in an advisory role.

Stakeholders should gain consensus among the counties and a representative number of municipalities within the proposed region. The process for gaining consensus may include a public officials' meeting(s).

In addition, the stakeholders should develop the management plan for the region. This plan should detail, at a minimum, the following issues:

1. The establishment of a Regional Trust, multi-municipal agreement or other arrangement acceptable to the Department, who would be responsible for implementation of the Wayfinding Program.
2. Establishment of an administrator (individual or entity) who would coordinate installation/maintenance/management of signing in the region.

3. Establishment of local accounts in the name of the entity created in item 1 above for funds collected from the participants for maintenance responsibilities.
4. Installation/maintenance responsibilities. Each PennDOT district office and county maintenance facility should be informed of the manager of the region, who would be responsible for maintenance of signs. Any problems with the sign (knockdown, vandalism) which are reported to PennDOT can be redirected to the appropriate manager. Stakeholders must establish emergency and routine response times for such situations.
5. Establishment of Program Guidelines, which should be in substantial conformance with the Department's Prototype Wayfinding Signing System Guidelines.
6. Development of a regional icon/logo and header color design.
7. Draft of a Trust Agreement, multi-municipal agreement or document of other arrangement developed in item 1 above.
8. Establishment of a budget and pro forma with documentation of the projected number of participants, program participation and maintenance fees and funding sources. Total estimated costs for administration, maintenance and implementation of signing. Provide supporting documentation to substantiate projections and estimates.
9. Determine necessary adjustment for existing signs (both TODS and other existing official traffic control devices).
10. Determine the approximate location and number of signs to be installed.

This plan must follow the specific rules set by PennDOT or MUTCD (whichever is more restrictive) for attractions to provide signs in the region.

PennDOT shall review and approve the Final Plan, considering consistency with any existing Regional Programs, program management and pro forma, and benefits to regional wayfinding.

D. Implementation

Following approval of the Final Plan, the Trust, multi-municipal agreement or other instrument may be formally created and the Wayfinding Program may be implemented in accordance with the Department-approved Program Guidelines and fee structure.

Engineering Construction Drawings should be prepared and approved by the Department for each sign installation.

The submission should include a scaled, sealed plan sheet developed generally in accordance with DM-3, indicating proposed sign structures along with existing PennDOT signs (regulatory, advisory, and warning) within 600 feet of each proposed sign structure. Specifications and bid documents should also be prepared and submitted to PennDOT for approval. These documents should closely conform to those normally used on Department let projects.

Part of the sign placement approval will be based on the location of existing signs. The proposed sign locations may be integrated into the Department's statewide GIS based sign inventory subsequent to installation. The information must include at a minimum the exact sign locations, sign type, approval date and installation date, and digital photographs of the sign taken from 10 and 150 feet. PennDOT district office information should also be included (district, county, route, segment, and offset). While determining the exact placement of the sign structures, the existing nearby signs will be located. The information on existing signs will be included in the statewide GIS if they are not there already.

After PennDOT approves the engineering construction drawings, specifications and bid documents, the bidding, contracting, and installation processes may begin.

V. Expansion of an Existing Signing Region

Should adjacent municipality(ies) outside the region desire to join the region after the region's inception, they submit a written request to the administrator and PennDOT and provide compelling reasons for their inclusion. The Department and administrator should thoroughly review the application and the compelling reasons to join the Region and render a decision. Should the application be granted, the potential participants from the adjacent municipality(ies) must, at a minimum, fund the relocation of gateway signs from the current entrance to the region to new region boundary.

VI. Contacts

Questions concerning any information contained within this toolbox may be directed to the appropriate Engineering District Office Traffic Unit. Listed below are the contact addresses and telephone numbers. The appropriate district office is based on the county as indicated in the table.

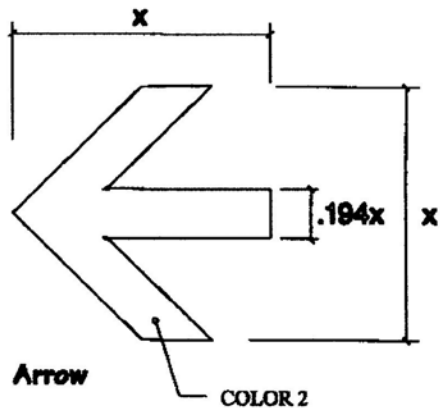
District	Address	Counties	Telephone
1-0	225 Elm Street Oil City, PA 16301	Erie, Warren, Forest, Venango, Mercer, Crawford	814-678-7085
2-0	1924-30 Daisy Street Clearfield, PA 16830	McKean, Potter, Elk, Cameron, Clinton, Clearfield, Centre, Mifflin, Juniata	814-765-0400
3-0	Jordon Avenue Montoursville, PA 17754	Tioga, Bradford, Lycoming, Sullivan, Montour, Union, Snyder, Northumberland, Columbia	570-368-8686
4-0	O'Neill Highway Dunmore, PA 18512	Susquehanna, Wayne, Luzerne Wyoming, Lackawanna, Pike,	570-963-4061
5-0	1713 Lehigh Street Allentown, PA 18103	Monroe, Carbon, Schuylkill, Berks, Lehigh, Northampton	610-798-4100
6-0	7000 Geerdes Blvd King of Prussia, PA 19406	Bucks, Montgomery, Chester, Philadelphia, Delaware	610-205-6700
8-0	2140 Herr Street Harrisburg, PA 17103	Adams, York, Franklin, Perry, Cumberland, Dauphin, Lebanon, Lancaster	717-787-6653
9-0	1620 N Juniata Street Hollidaysburg, PA 16648	Blair, Huntingdon, Fulton, Somerset, Bedford, Cambria	814-696-7250
10-0	P.O. Box 429 Indiana, PA 15701	Butler, Indiana, Armstrong, Clarion, Jefferson	724-357-2800
11-0	45 Thoms Run Road Bridgeville, PA 15017	Lawrence, Allegheny, Beaver	412-429-5000
12-0	P.O. Box 459 Uniontown, PA 15401	Greene, Fayette, Washington, Westmoreland	724-439-7315

Typeface 1/Friz Quadrata

COLOR 2

Typeface 2/Friz Quadrata Bold

COLOR 2



Color 1 - 3M Translucent Film #3650-157 European Blue

Color 2 - 3M Scotchlite Diamond Grade #980-10 White

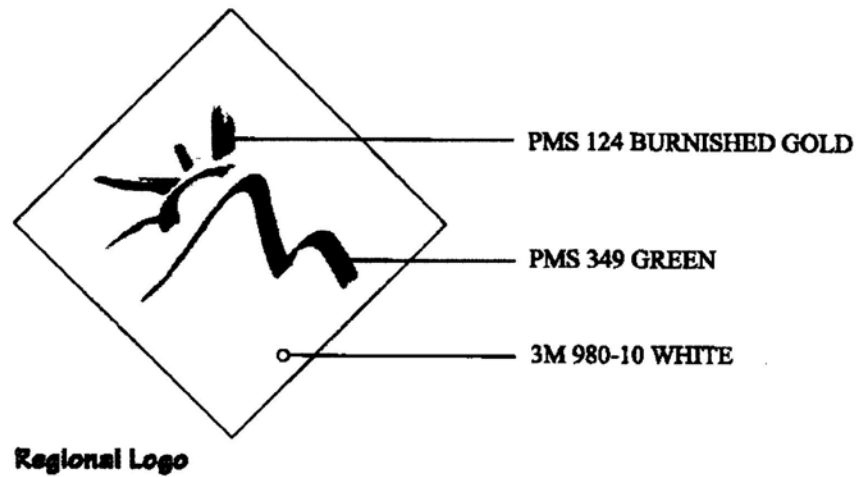


Figure 1.15.1

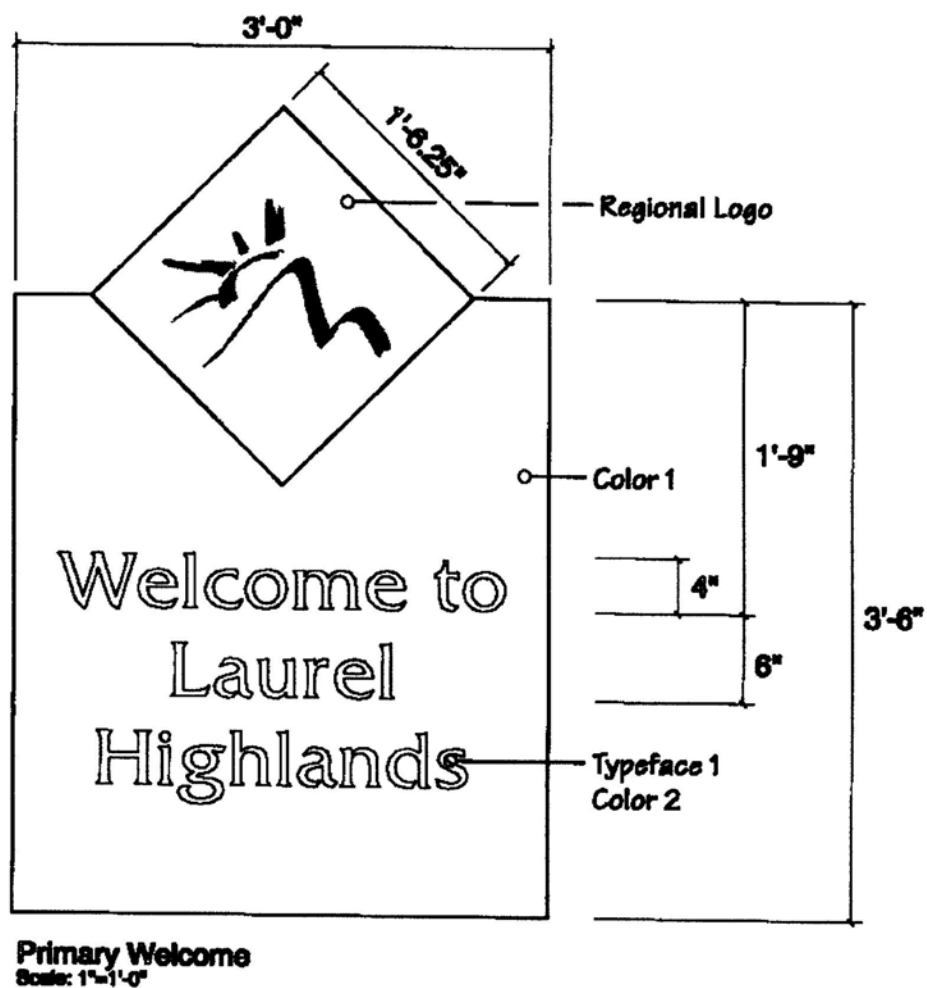


Figure 1.15.2

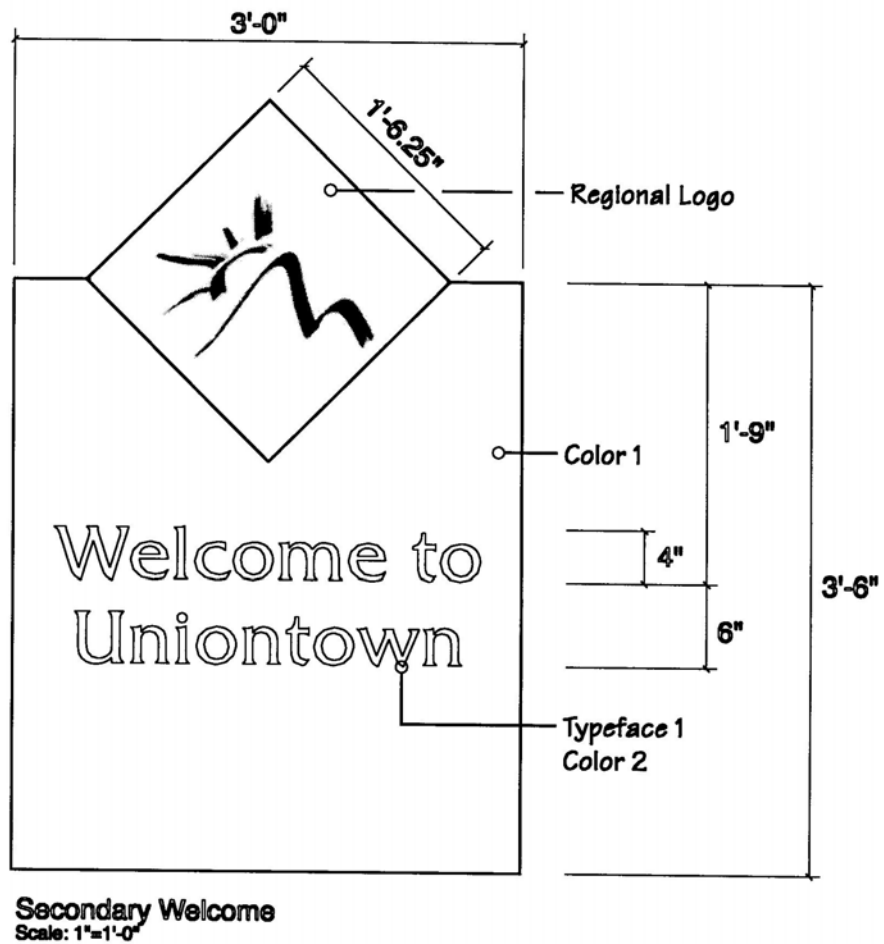


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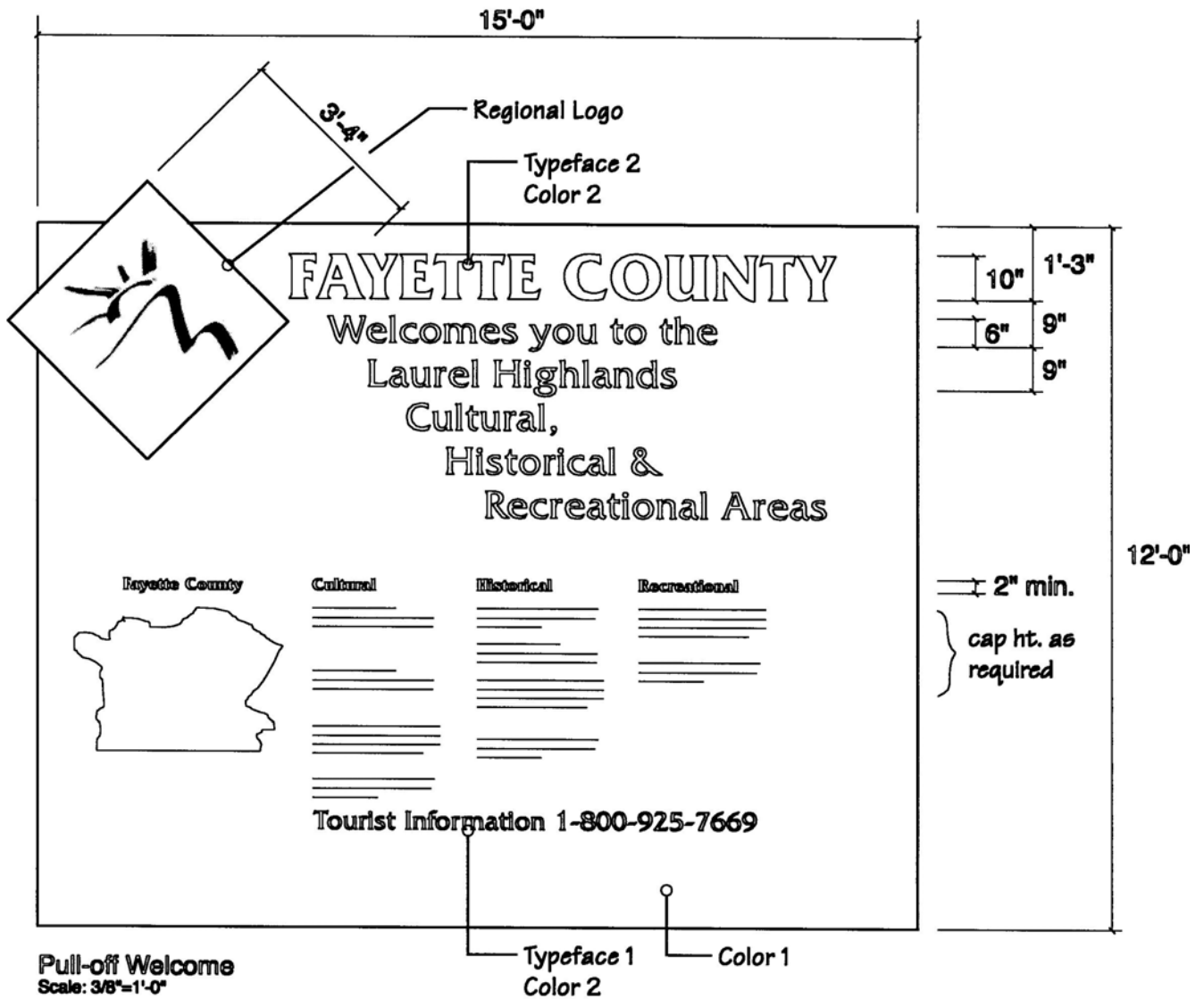


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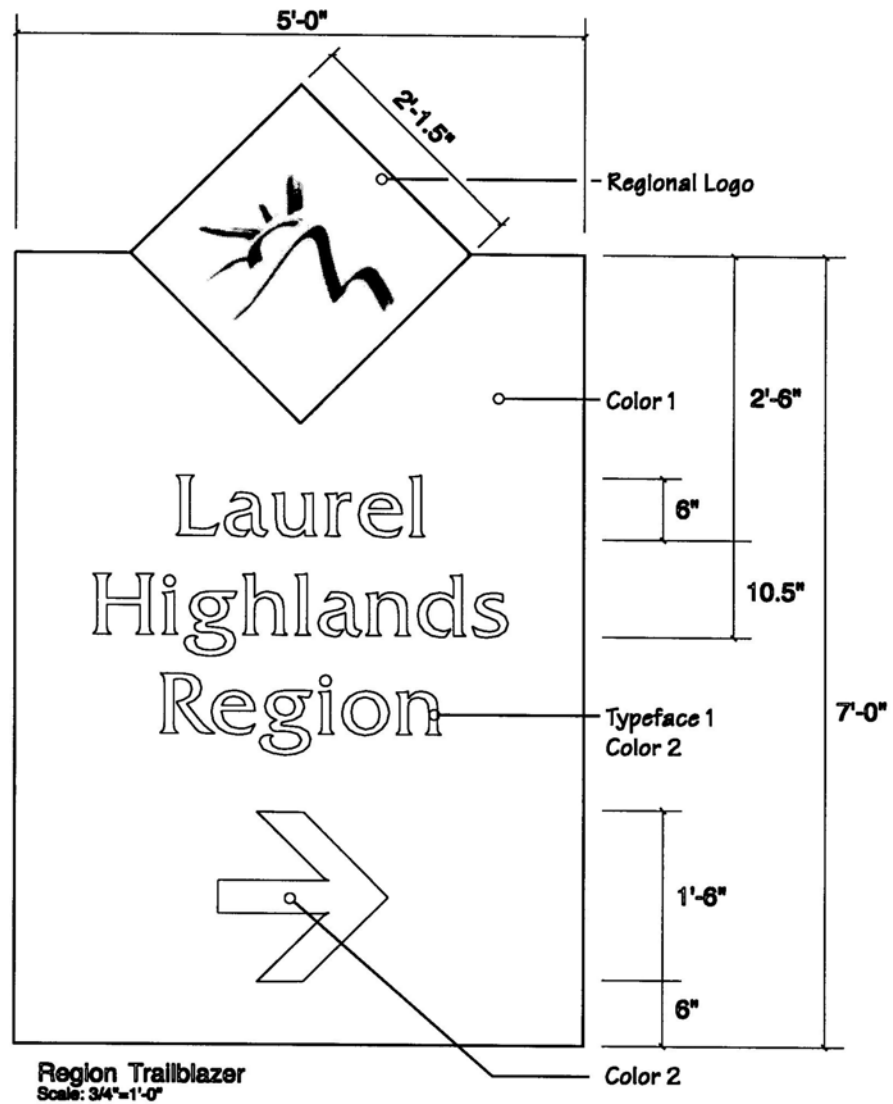


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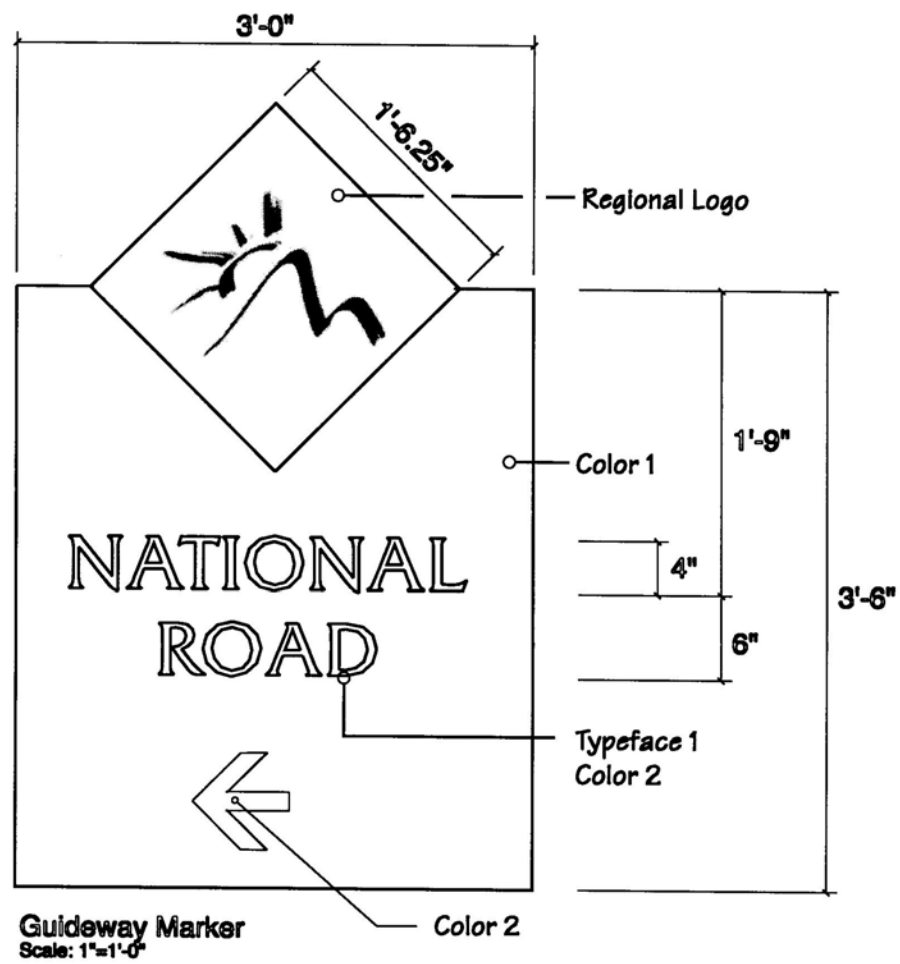
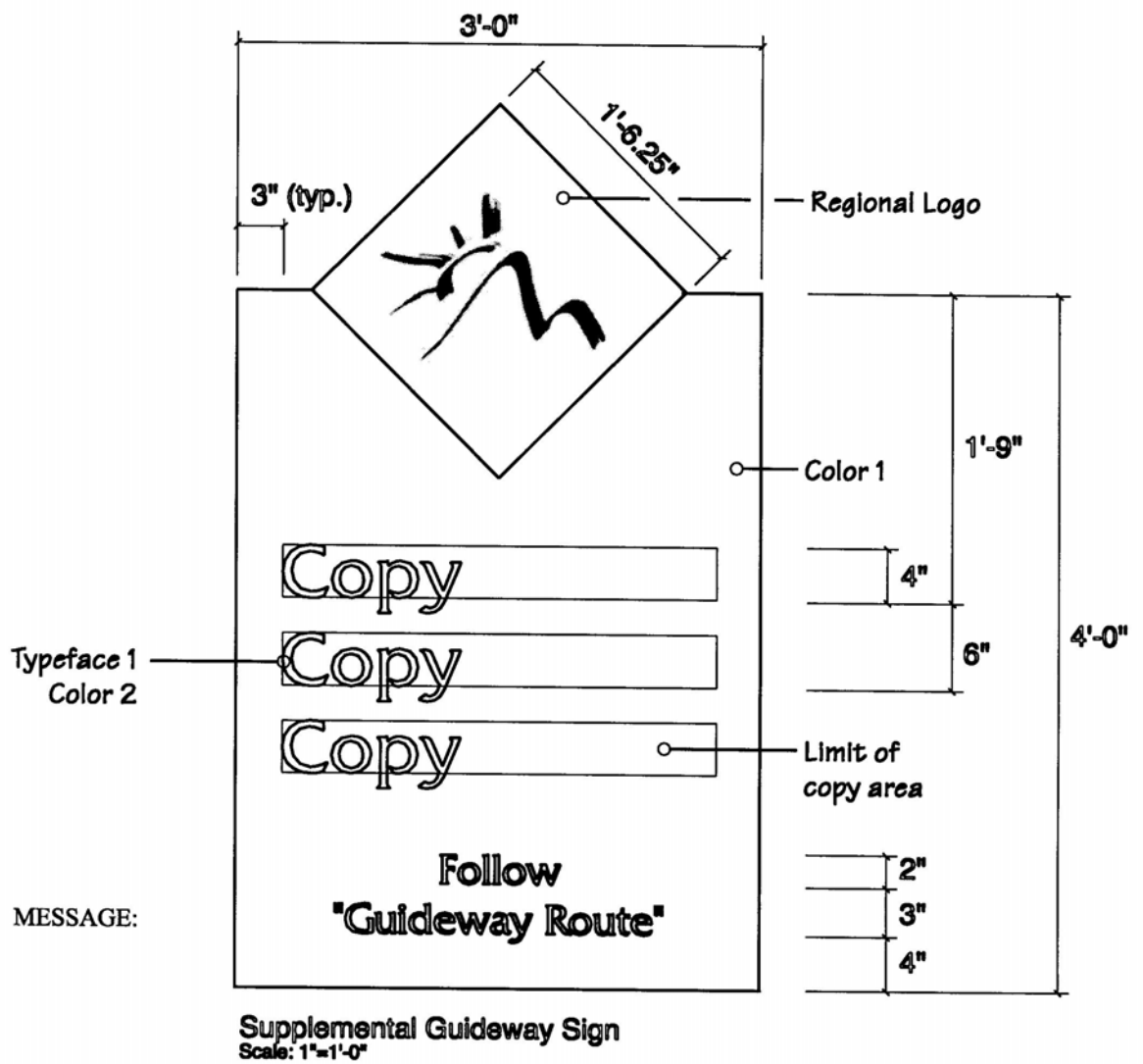
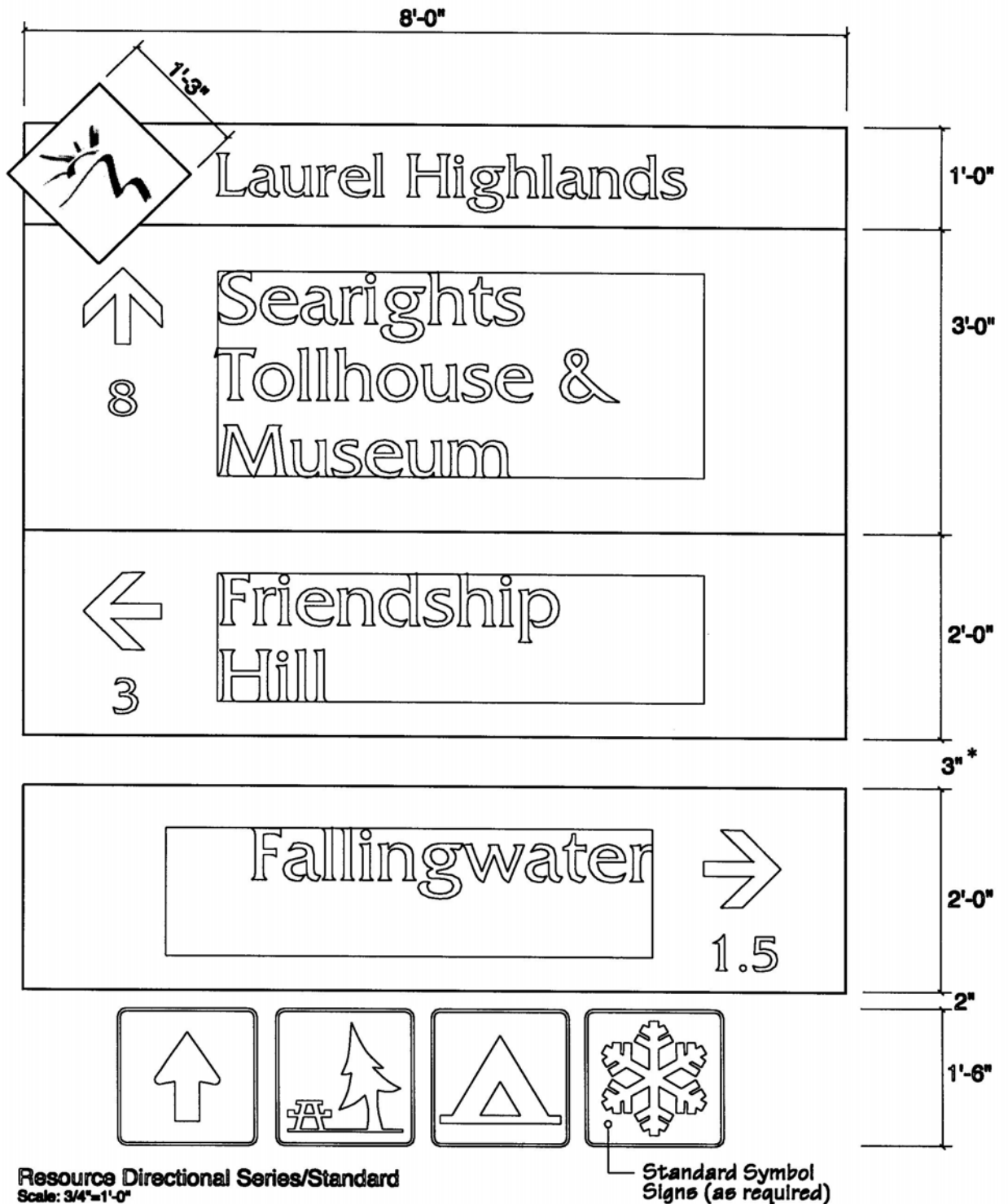


Figure 1.15.6



NOTE: MESSAGE MAY VARY.....NEXT LEFT/RIGHT ETC.

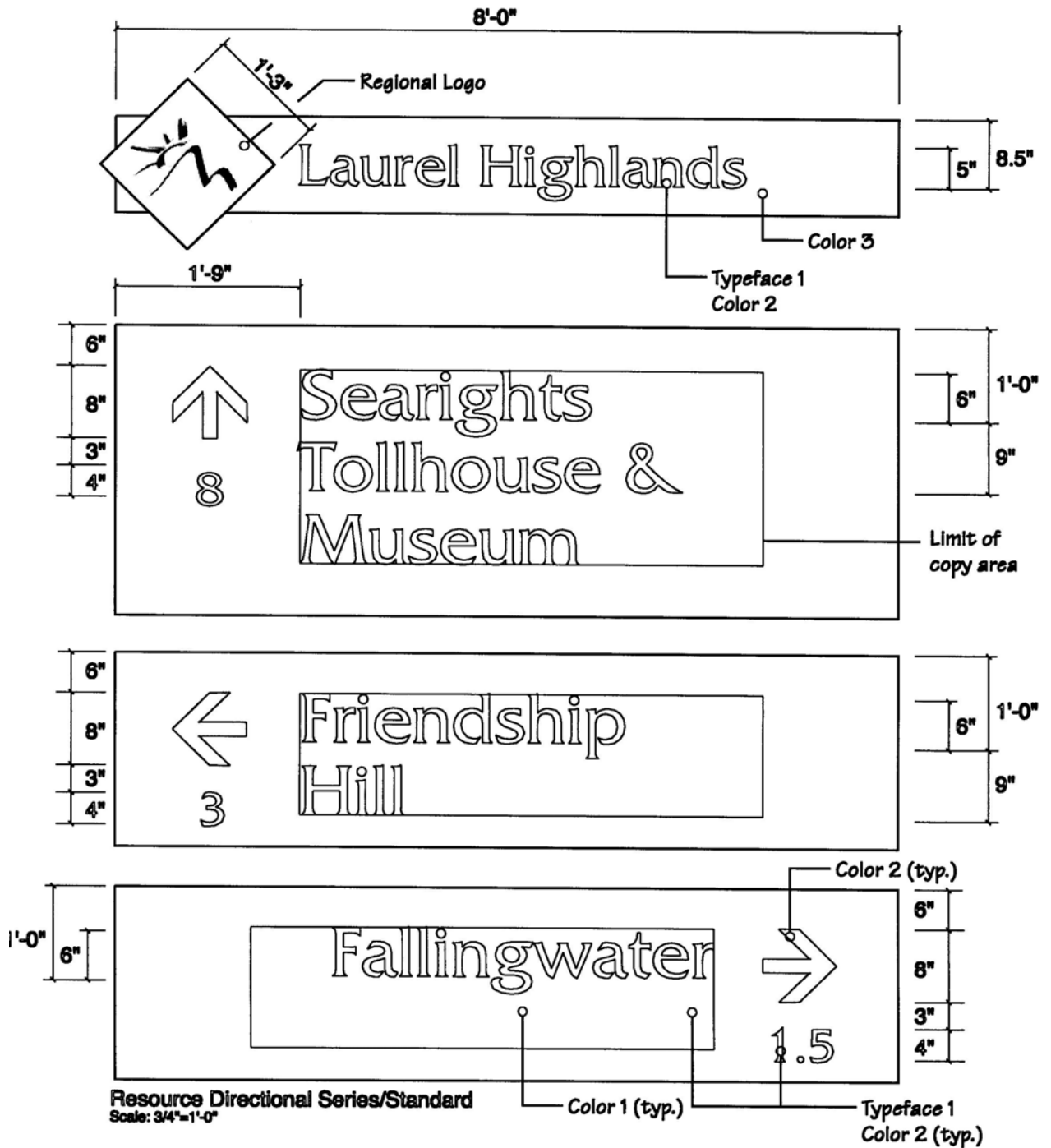
Figure 1.15.7



"STANDARD" FOR USE ON ROADS WITH A POSTED SPEED LIMIT GREATER THAN 40 MPH.
LEGEND HEIGHT 6"

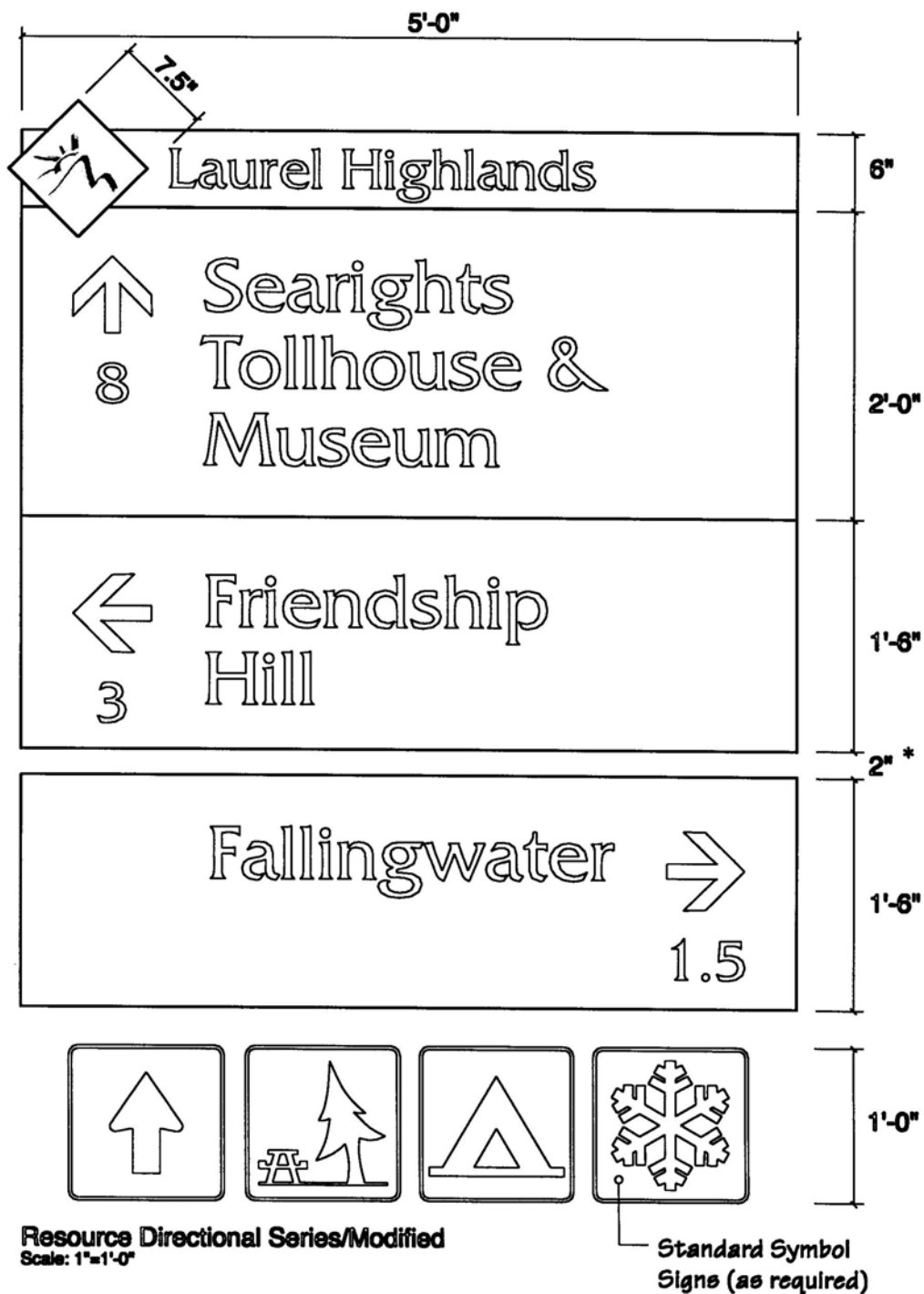
* USE SPACE BETWEEN PANELS WITH LEFT-JUSTIFIED ARROWS AND PANELS WITH RIGHT-JUSTIFIED ARROWS.

Figure 1.15.8a



"STANDARD" FOR USE ON ROADS WITH A POSTED SPEED LIMIT GREATER THAN 40 MPH.
LEGEND HEIGHT 6"

Figure 1.15.8b

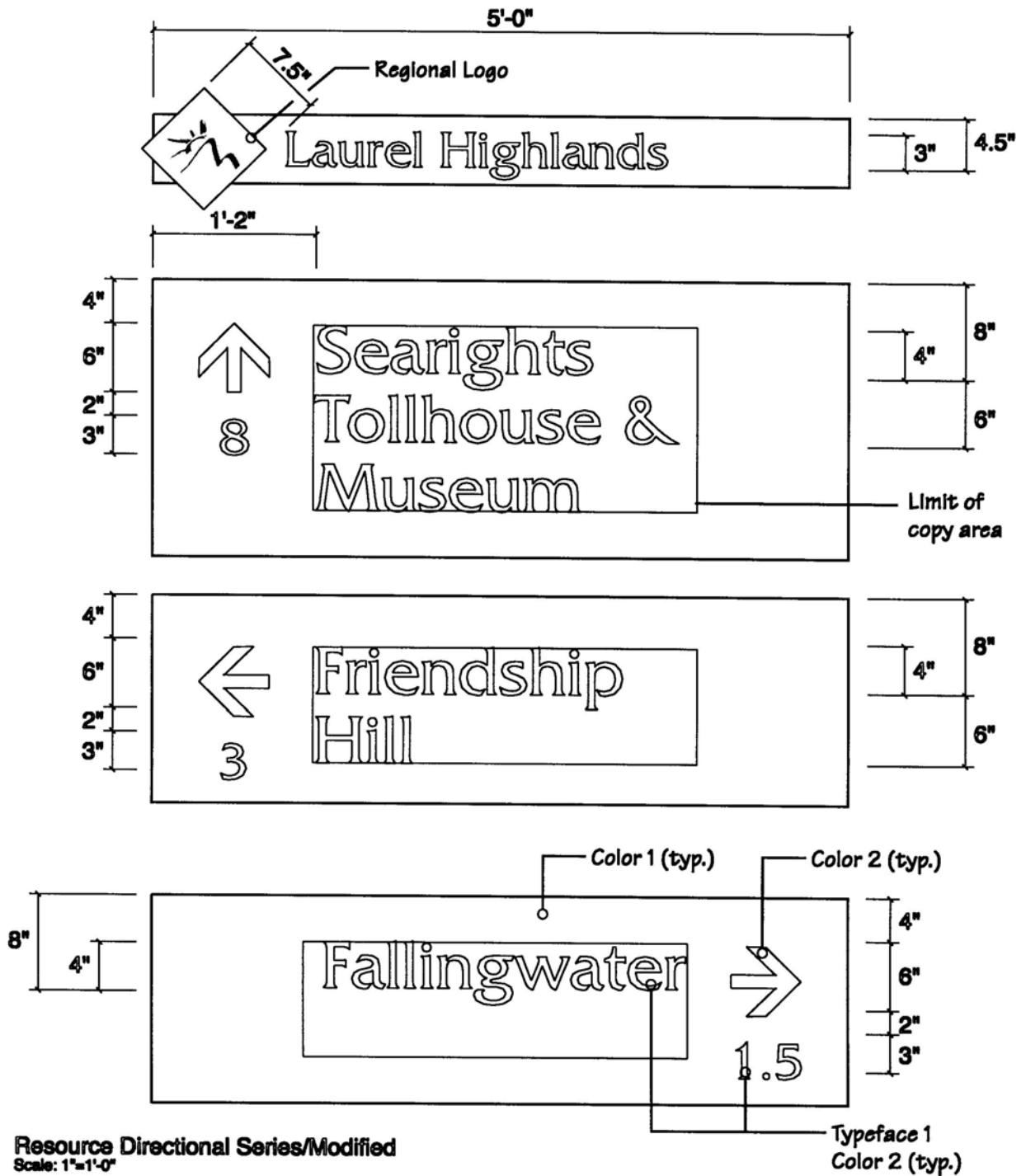


"MODIFIED" FOR USE ON ROADS WITH A POSTED SPEED LIMIT OF 40 MPH OR UNDER, OR WHERE RIGHT-OF-WAY IS INSUFFICIENT FOR THE STANDARD SIGN.

LEGEND HEIGHT 4"

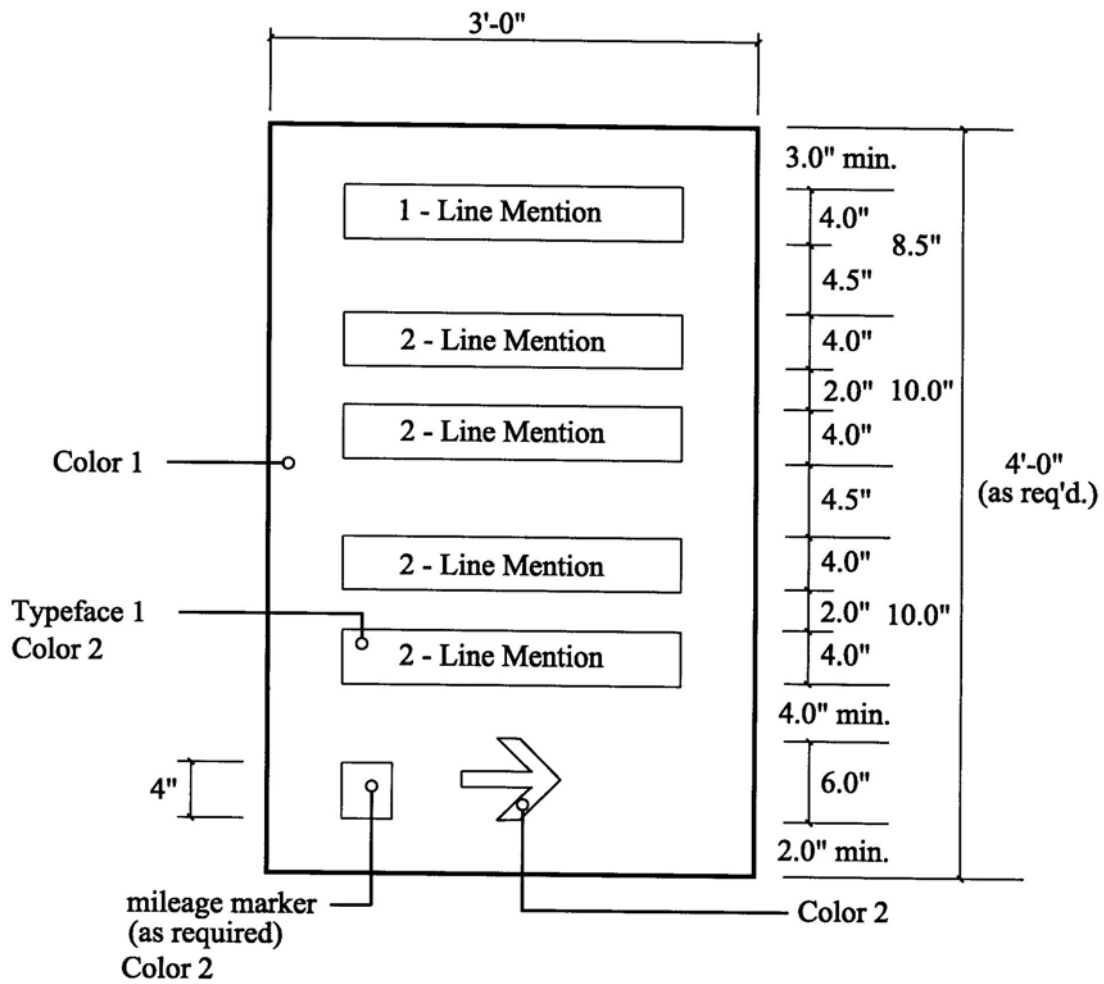
- * USE SPACE BETWEEN PANELS WITH LEFT-JUSTIFIED ARROWS AND PANELS WITH RIGHT-JUSTIFIED ARROWS.

Figure 1.15.9a



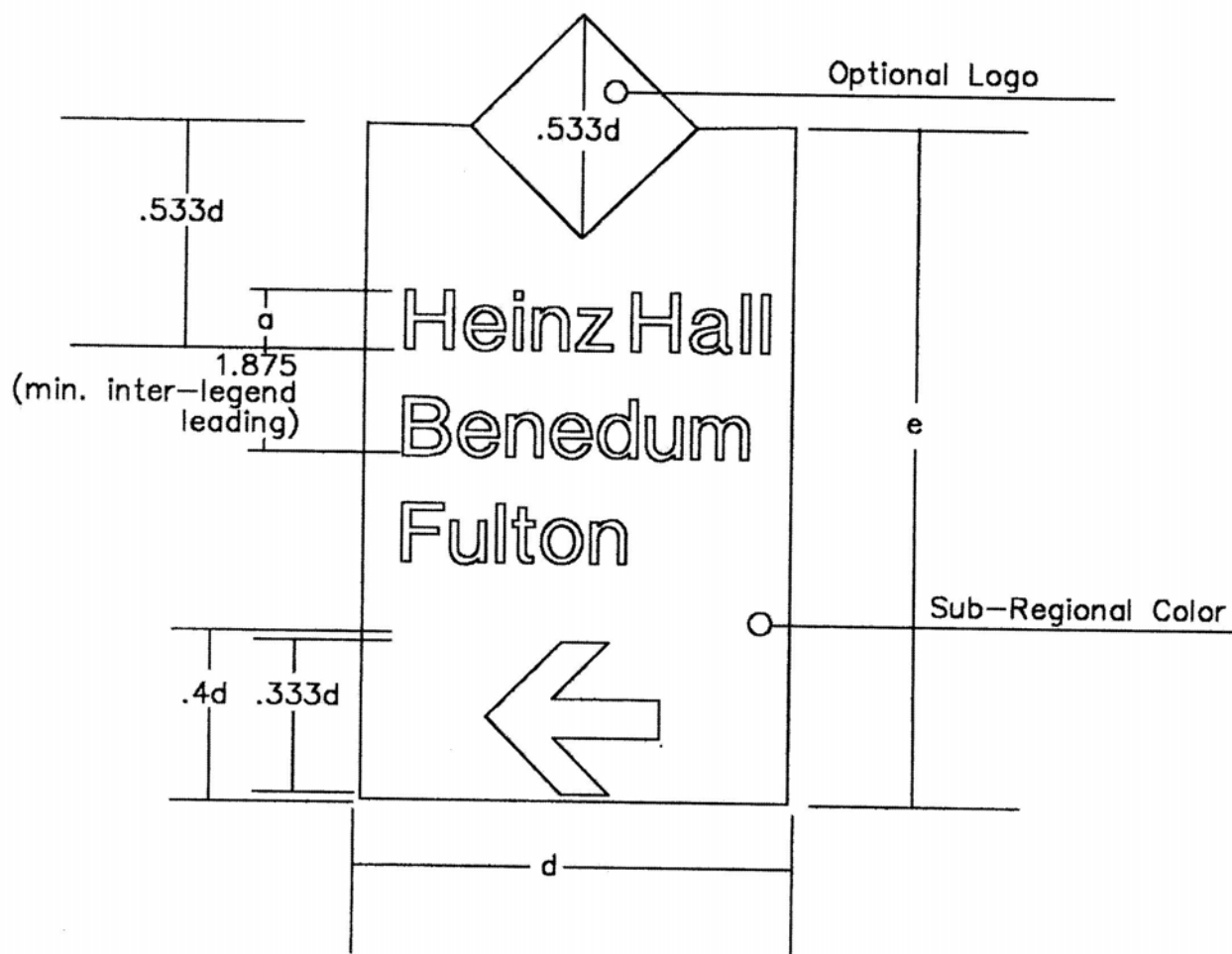
"MODIFIED" FOR USE ON ROADS WITH A POSTED SPEED LIMIT OF 40 MPH OR UNDER,
 OR WHERE RIGHT-OF-WAY IS INSUFFICIENT FOR THE STANDARD SIGN.
 LEGEND HEIGHT 4"

Figure 1.15.9b



Resource Directional Series/Vertical
Scale: 1"=1'-0"

Figure 1.15.10



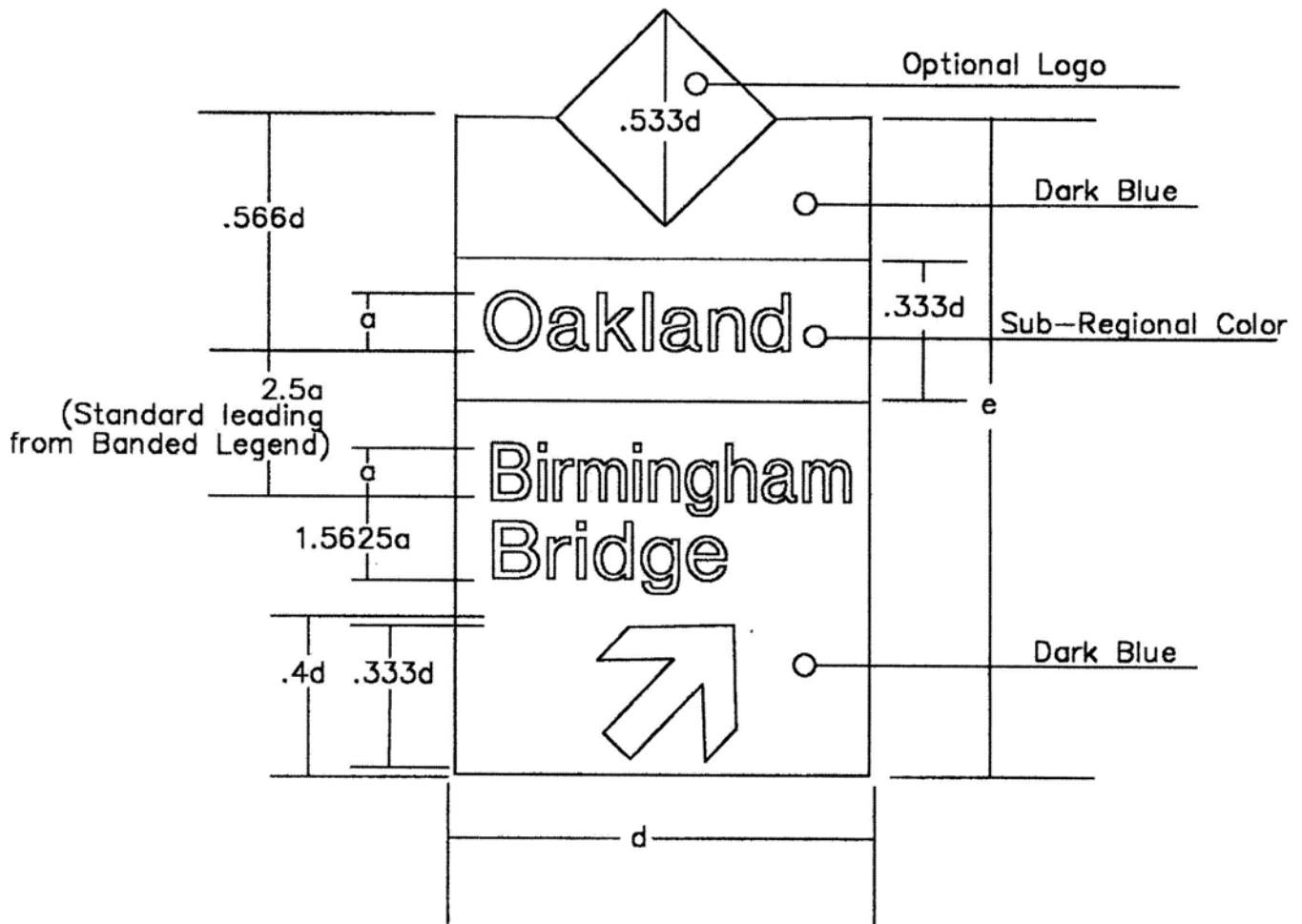
3-Line Sign Height

a = legend cap height	4"	4.8"	6"
b = sub-legend cap height (not shown)	3.5"	4.2"	5.25"
c = sub-sub-legend cap height (not shown)	x	3.6"	4.5"
d = sign width	30"	36"	48"
e = sign height (not including logo)	48"	57.6"	72"

5 - Line Sign Height

e = sign height	60"	72"	90"
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Figure 1.15.11



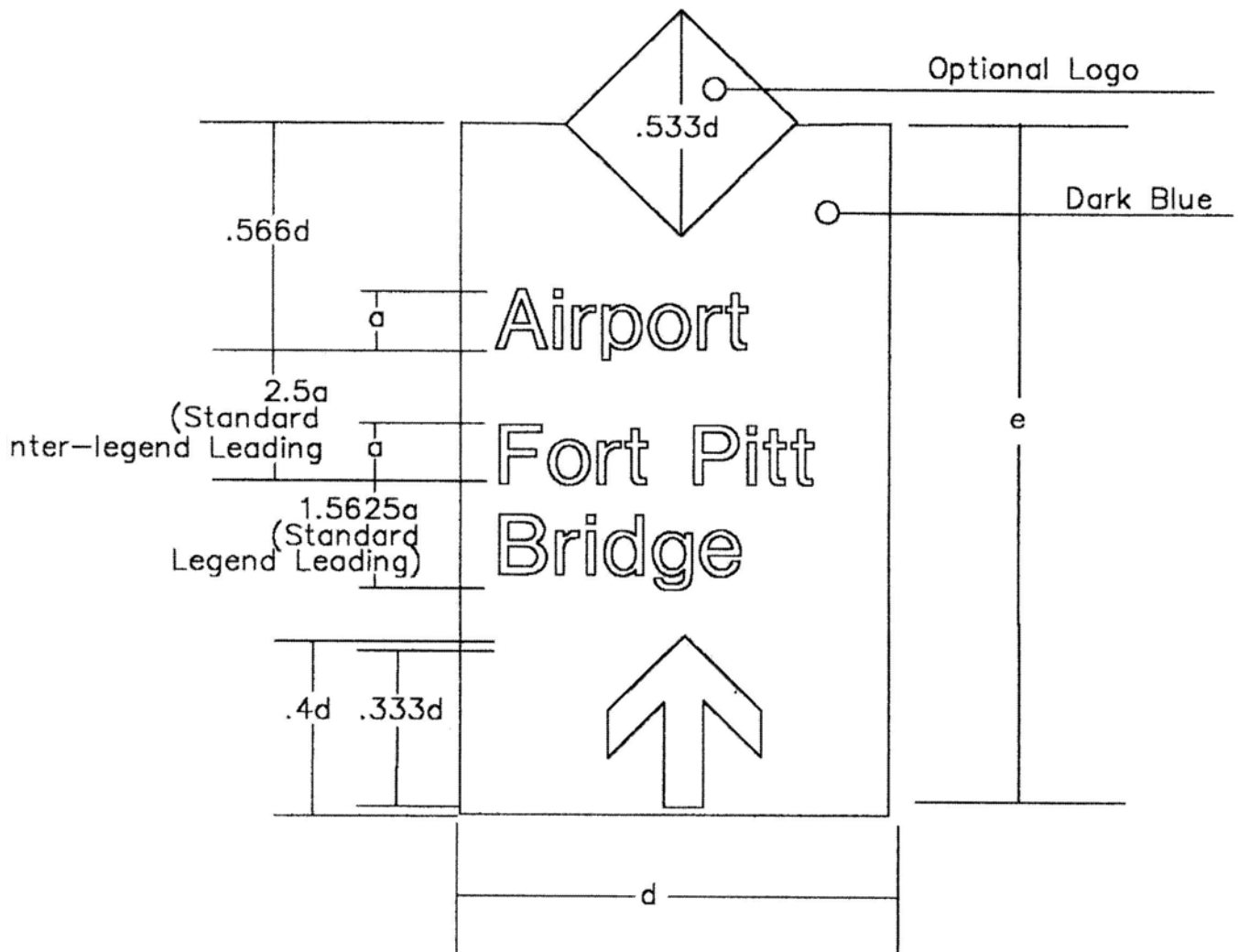
3-Line Sign Height

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5 - Line Sign Height

e = sign height	60"	72"	90"
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Figure 1.15.12



3-Line Sign Height

a = legend cap height	4"	4.8"	6"
b = sub-legend cap height (not shown)	3.5"	4.2"	5.25"
c = sub-sub-legend cap height (not shown)	x	3.6"	4.5"
d = sign width	30"	36"	48"
e = sign height (not including logo)	48"	57.6"	72"

5 - Line Sign Height

e = sign height	60"	72"	90"
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Figure 1.15.13